

Generation Interconnection System Impact Study Report

For Queue Project Q20-1580

Villa Del Rey 3010 138 kV

September 26, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1580. The Interconnection Customer proposes a 144.4 MW wind facility interconnecting at the Villa Del Rey 3010 138 kV substation in the Virginia area. The request was received on October 01, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 391 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 19 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,862,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
N Randolph / Bairoa 3008 138 KV	Fritch / Big Hanaford 138 KV	118.5	237	6.1
Chpola / Strathmore 138 KV line	Fritch / Lakeside 138 KV	111.4	285	6.1
Riv Road / Humacao 2601 138 KV line	Lakeside / Fritch 138 KV	111.0	341	6.1
Riv Rd / Fritch Substation 138 KV	Lakeside Substation / Fritch 138 KV	109.2	368	6.1
Westcoast Cellufibre / Leavitt 138 KV Circuit 1	Mitchell Dam Substation / Skaggs Island Jct 2 345 KV line	108.1	384	9.7
Westcoast Cellufibre / Leavitt 138 KV Ckt 2	Mitchell Dam / Skaggs Island Jct 2 Substation 345 KV line	108.1	384	9.7
Strathmore / Humacao 2601 138 KV	Fritch / Lakeside 138 KV	107.3	323	6.1
Carver / Big Hanaford 138 KV	Plaza / Big Hanaford 138 KV Line	106.7	495	7.4
N Randolph / Yellow Crk 138 KV	Fritch / Big Hanaford 138 KV	105.8	279	6.1
Kings Riv / Dg Amp Solar Coldwater 138 KV	Big Hanaford Substation / Frtch 138 KV	105.4	435	6.9
Caver / Shockoe 138 KV	Plaza / Big Hanaford 138 KV	103.8	581	7.4
Fritch / Big Hanaford 138 KV	Mitchell Dam / Skaggs Island Junction 2 345 KV line	103.6	854	13.1
Natchez / Chancellor Row 138 KV	Skaggs Island Jct 2 / Mitchell Dam 345 KV	102.6	508	8.9
Spring St / Chipola 138 KV Line	Mitchell Dam / Skaggs Island Jct 2 345 KV	102.5	234	8.0
Lakeside / Natchez Substation 138 KV L	Mitchell Dam / Skaggs Island Jct 2 345 KV	102.2	531	8.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Northern Neck 345-138 kV Transformer Bank	Chancellor Row / Skaggs Island Jct 2 Substation 345 KV	101.4	637	6.0
Frith / Lakeside 138 KV	Mitchell Dam / Skaggs Island Jct 2 345 KV	101.1	607	8.9
Stepdown / Hardy Reit 138 KV	Yarnen / Mira Vista 345 KV	100.9	560	5.6
Haggin 345/138 kV Transformer Bank	N Tower Tap / Haggin 345 KV	100.5	426	8.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
N Randolph / Bairoa 3008 138 KV	Rebuild line with higher-capacity conductor	\$11,216,000	2% (\$278,000)
Chipola / Strathmore 138 KV	Rebuild line with higher-capacity conductor	\$1,963,000	2% (\$39,000)
Riv Road / Humacao 2601 138 KV line	Rebuild line with higher-capacity conductor	\$2,405,000	2% (\$47,000)
Fritch / River Rd Substation 138 KV Line	Reconductor line	\$860,000	2% (\$17,000)
Westcoast Cellufibre / Leavitt 138 KV Circuit 1	Reconductor line	\$564,000	2% (\$10,000)
Leavitt / Westcoast Cellufibre 138 KV Cir. 2	Reconductor line	\$675,000	2% (\$12,000)
Strathmore / Humacao 2601 138 KV	Reconductor line	\$781,000	2% (\$15,000)
Big Hanaford / Carver Substation 138 KV	Reconductor line	\$591,000	2% (\$12,000)
N Randolph / Yellow Crk 138 KV	Reconductor line	\$2,979,000	2% (\$74,000)
Kings River / Dg Amp Solar Coldwater 138 KV	Reconductor line	\$1,145,000	2% (\$19,000)
Caver / Shockoe 138 KV	Reconductor line	\$552,000	2% (\$11,000)
Fritch / Big Hanaford 138 KV line	Reconductor line	\$1,740,000	2% (\$29,000)
Chancellor Row / Natchez 138 KV	Reconductor line	\$888,000	2% (\$16,000)
Spring Street / Chipola 138 KV line	Reconductor line	\$2,861,000	3% (\$83,000)
Lakeside / Natchez 138 KV	Reconductor line	\$521,000	2% (\$9,000)
Northern Neck Substation 345 kV/138 kV Transformer Bank	Replace transformer with higher-rated unit	\$16,428,000	3% (\$511,000)
Fritch / Lakeside Substation 138 KV	Reconductor line	\$916,000	2% (\$16,000)
Stepdown / Hardy Reit 138 KV	Reconductor line	\$3,407,000	3% (\$96,000)
Haggin 345 kV/138 kV XFMR	Replace transformer with higher-rated unit	\$22,267,000	2% (\$568,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
	Total allocated to project		\$1,862,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.